



DIGITAL LOGIC

Chapters 6 and 7

Analysis and Design of Asynchronous Logic Circuits

MSI General-purpose Circuits

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Chapter 7 MSI

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The development of integrated circuits from SSI to MSI, LSI and VLSI has made the logic functions of a single chip more and more powerful.

In general, SSI is the integration of basic devices (such as logic gates or flip-flops)

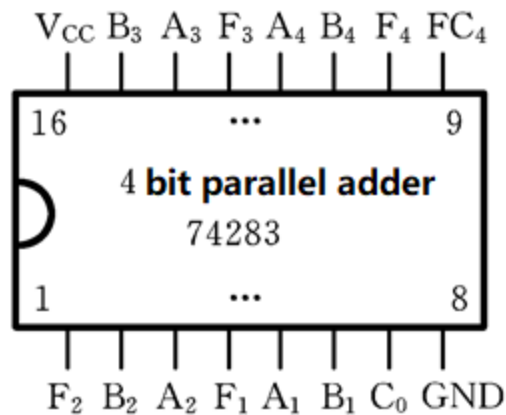
MSI is already an integration of logic components (such as decoders, registers, etc.) ;

However LSI and VLSI is the Integration of digital subsystems or entire digital systems such as microprocessors.

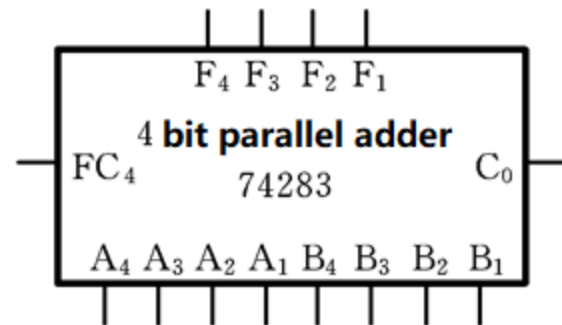
The digital system composed of medium and large scale integrated circuits has the advantages of small size, low power consumption, high reliability, and is easy to design, debug and maintain.

7.1 Common medium-scale combinational logic circuit

The most widely used medium-scale combinatorial logic integrated circuits are binary parallel adders, decoders, encoders, multiplexers, demultiplexers, and registers.



(a)



(b)

where, A_4, A_3, A_2, A_1 ----- Binary addends;
 B_4, B_3, B_2, B_1 ----- Binary summand;
 F_4, F_3, F_2, F_1 ----- Sum;
 C_0 ----- Carry from lower bit;
 FC_4 ----- Carry to the higher bit.





Applications:

In addition to binary addition, the binary parallel adder can also implement functions such as transcoding, binary subtraction, binary multiplication, and decimal addition.



Example A 4-bit binary parallel adder/subtractor is designed with a 4-bit binary parallel adder.

Solution

Analysis: According to the problem requirements, set the subtraction method to use the 2s complement code; so we can define as

$A = a_4a_3a_2a_1$ ----- Addend(or Minuend);

$B = b_4b_3b_2b_1$ ----- Summand(or subtrahend);

$S = s_4s_3s_2s_1$ ----- Sum(or difference);

M -----is function selection variable. If $M=0$, execute $A+B$;
if $M=1$, execute $A-B$.

Conclude the following function through the 2s complement calculation rule :

If $M=0$, execute $a_4a_3a_2a_1 + b_4b_3b_2b_1 + 0$ ($A+B$)

If $M=1$, execute $a_4a_3a_2a_1 + \bar{b}_4\bar{b}_3\bar{b}_2\bar{b}_1 + 1$ ($A-B$)





The above logic function can be implemented by a 4-bit binary parallel adder and 4 XOR gates.

Details:

put 4-bit binary number $a_4a_3a_2a_1$ to the parallel adder's input $A_4A_3A_2A_1$, each of the 4 bit binary number $b_4b_3b_2b_1$ exor M, the values are passed to the input of $B_4B_3B_2B_1$; and pass M to C_0 .

M=0: $A_i=a_i, B_i=b_i, C_0=0$

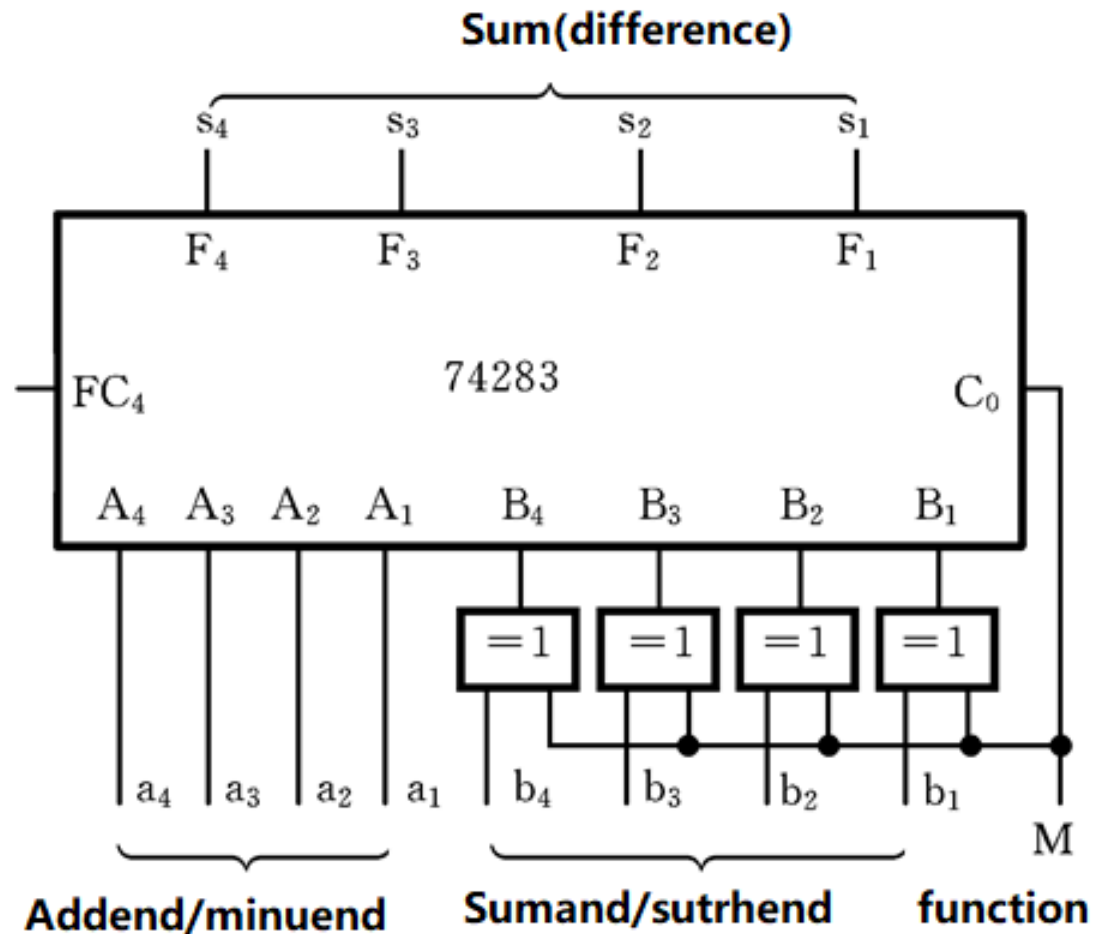
thus: $a_4a_3a_2a_1 + b_4b_3b_2b_1 + 0$ (A+B);

M=1: $A_i=a_i, B_i=\overline{b_i}, C_0=1$,

thus: $a_4a_3a_2a_1 + \overline{b_4b_3b_2b_1} + 1$ (A-B) .



The logic circuit diagram for implementing the given function is as follows:





7.1.2 Decoder and Encoder

Decoder and Encoder is a multi-input and multi-output combinational logic component widely used in digital systems. ◦

I. Decoder

The function of the decoder is to "translate" the input code with a specific meaning and convert it into a corresponding output signal.

There are many types of decoders, such as binary decoders, binary-decimal decoders, and digital display decoders.





1. Binary decoder

Binary decoder: A multi-output combinational logic circuit capable of transforming n input variables into 2^n output functions, in particular, the output function is corresponding to a minterm formed by the input variables.

(1) Features :

▲ A binary decoder typically has n inputs, 2^n outputs, and one (or more) enable inputs;

▲ When the enable input is at the active level, for each set of input codes, only one output is active and the remaining outputs are inactive.

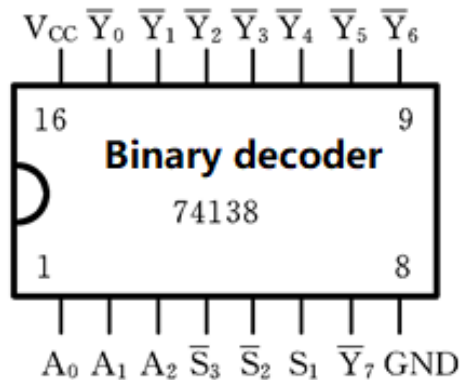
▲ The active level can be either high (called high level decoding) or low (referred to as low level decoding).



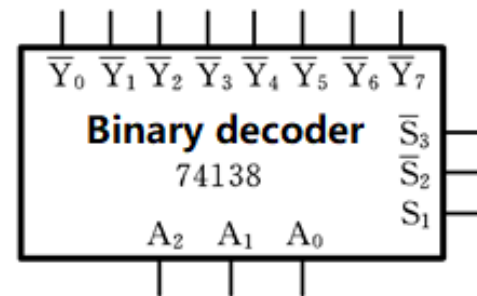
(2) Typical chips

Common MSI binary decoders have 2-4 lines (2 input 4 output) decoder, 3-8 line (3 input 8 output) decoder and 4-16 line (4 input 16 output) decoders, etc. .

Figures (a) and (b) show the pin arrangement diagram and logical symbols of the 74138 type 3-8 line decoder, respectively.



(a)



(b)

where, A_2 , A_1 , A_0 --- Inputs; $\overline{Y}_0 \sim \overline{Y}_7$ --- Outputs;
 $S_1, \overline{S}_2, \overline{S}_3$ --- Enable;

74138 Decoder truth table

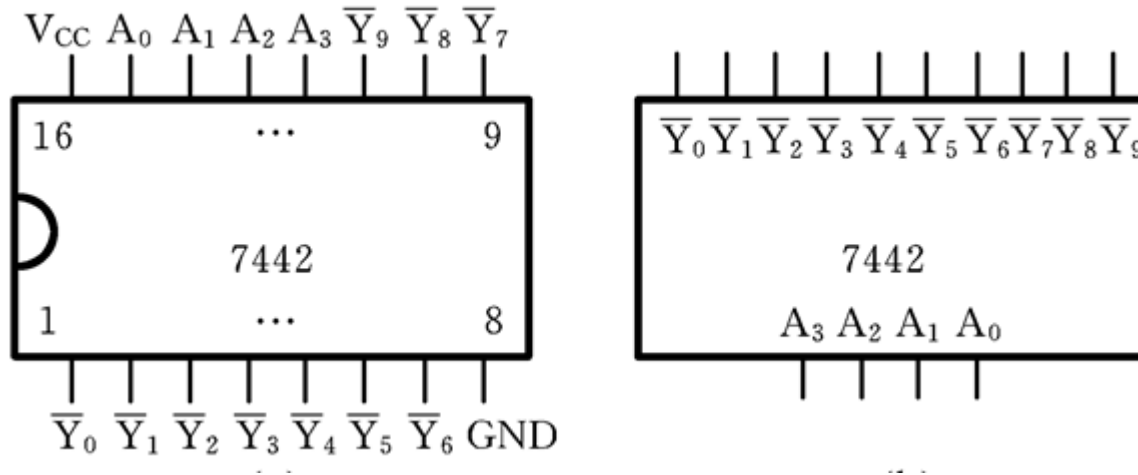
Inputs					Outputs							
S_1	$\overline{S_2} + \overline{S_3}$	A_2	A_1	A_0	$\overline{Y_0}$	$\overline{Y_1}$	$\overline{Y_2}$	$\overline{Y_3}$	$\overline{Y_4}$	$\overline{Y_5}$	$\overline{Y_6}$	$\overline{Y_7}$
1	0	0	0	0	0	1	1	1	1	1	1	1
1	0	0	0	1	1	0	1	1	1	1	1	1
1	0	0	1	0	1	1	0	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1	1	1
1	0	1	0	0	1	1	1	1	0	1	1	1
1	0	1	0	1	1	1	1	1	1	0	1	1
1	0	1	1	0	1	1	1	1	1	1	0	1
1	0	1	1	1	1	1	1	1	1	1	1	0
0	d	d	d	d	1	1	1	1	1	1	1	1
d	1	d	d	d	1	1	1	1	1	1	1	1

If $S_1 = 1, \overline{S_2} + \overline{S_3} = 0$, whatever are the values of A_2, A_1 and A_0 , the outputs $\overline{Y_0} \cdots \overline{Y_7}$ have only one level output (0), the rest are 1.

2. Binary-decimal decoder

Function: translate the 10 sets of code of 4 bit BCD codes into 10 decimal digit symbols corresponding to the output signal.

For example, a common chip 7442 is a decoder that converts 8421 codes into decimal numbers. The chip pin diagram and logic symbols are as follows.



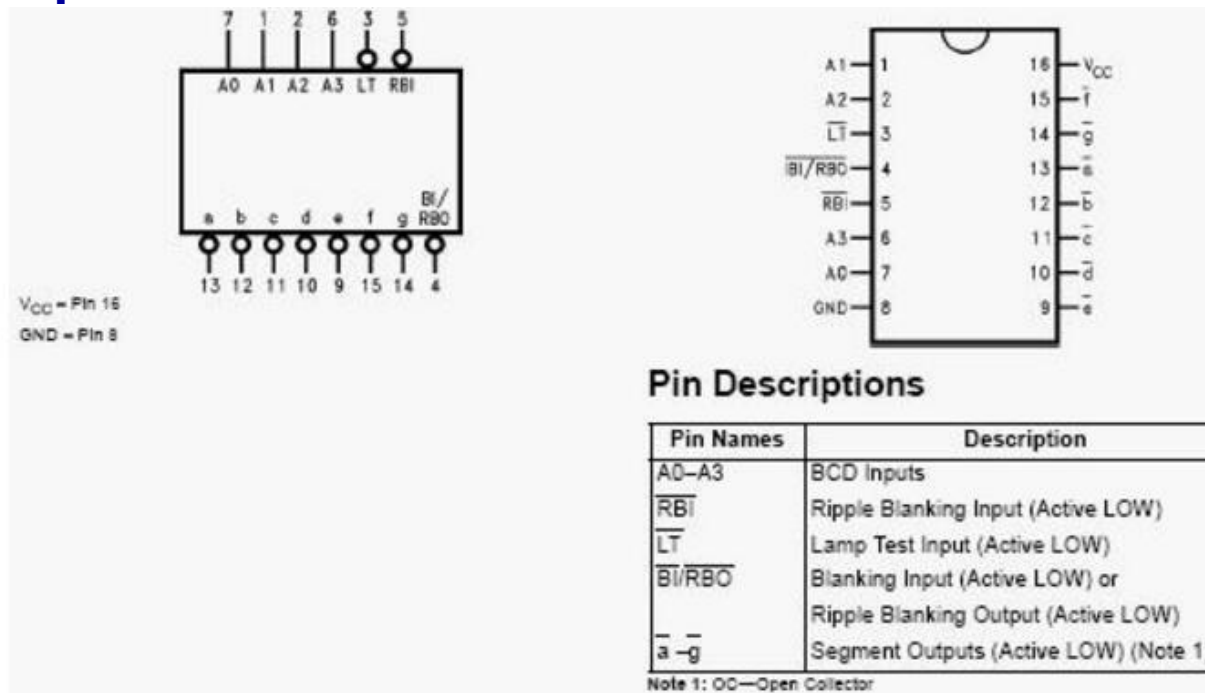
The output of the decoder is active low. Secondly, for the 6 illegal codes (1010~1111) that are not allowed in the 8421 code, the decoder outputs $\overline{Y}_0$ $\overline{Y}_9$ can not produce the low level signal, that is, the decoder refuses to translate the 6 illegal codes.

3. Digital display decoder

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Function: The digital display decoder is the core component of the driving display device (such as fluorescent digital tube, liquid crystal digital tube, etc.), which can convert the input code into corresponding numbers and display them on the digital tube.

Commonly used digital display decoder has seven segment digital display decoder and eight segment digital display decoder.
For example 74LS47





II. Encoder

Functions: The function of the encoder is exactly the opposite of the decoder. It is to arrange the input signals according to a certain rule, so that each group of output codes has its specific meaning.

Types: The encoder has various types according to the different characteristics and requirements of the signal to be encoded. The most common ones are binary-decimal encoders (also known as decimal-BCD code encoders) and priority encoders.

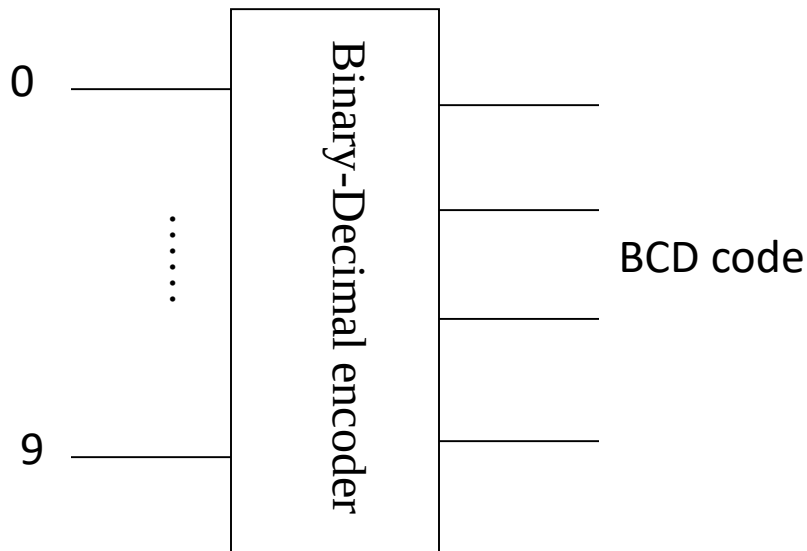
1. Binary-Decimal encoder

(1) **Function:** Encode decimal numbers 0 to 9 into 4-bit BCD codes.



(2) Structure diagram.

This encoder represents 10 different numbers from 10 inputs and 4 outputs represent the corresponding BCD code. The block diagram is as follows:



Note: The input signals of the binary-decimal encoder are mutually exclusive, that is, only one input is allowed to be a valid signal at any time.





7.1.3 Multiplexer and demultiplexer

Multiplexers and demultiplexers are medium scale integrated circuits commonly used in digital systems. Its basic function is to complete the selection and distribution of multiple data, and realize the time-sharing transmission of multiple data on the common transmission line. In addition, various logic functions such as parallel-to-serial conversion of data, generation of sequence signals, and various logic function functions can be performed.

I. Multiplexer

Multiplexers also known as data selectors or multiplexers, are commonly referred to as MUX. It is a combinational logic circuit with multiple inputs and single outputs.

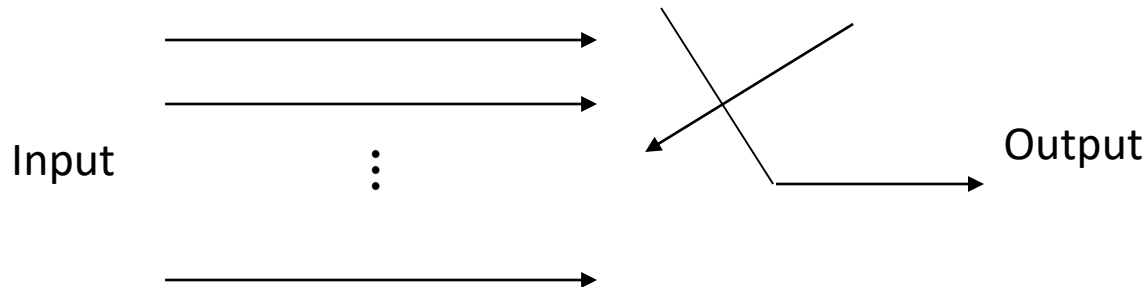


1. Logical characteristics

(1) Logic function: Select one of the multiple inputs to send to the output, and the output selection for the input is controlled by the selected control amount. Generally, a multiplexer having 2^n inputs and one output has n selection control variables, and each combination of control variables is selected to be sent to the output corresponding to one input.

(2) Basic Idea

The multiplexer is constructed as a single-pole multi-throw switch, ie

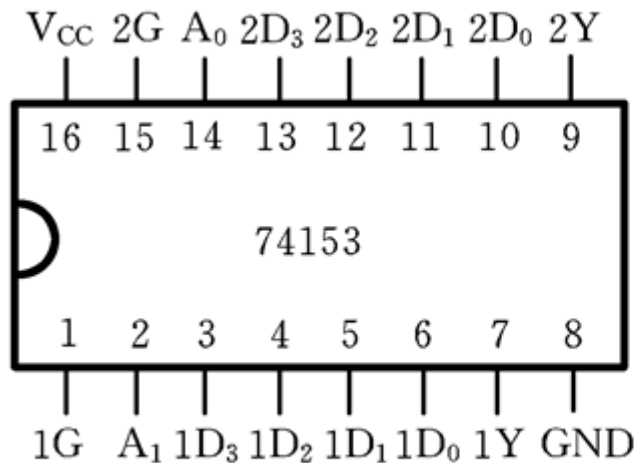


2. Typical chips

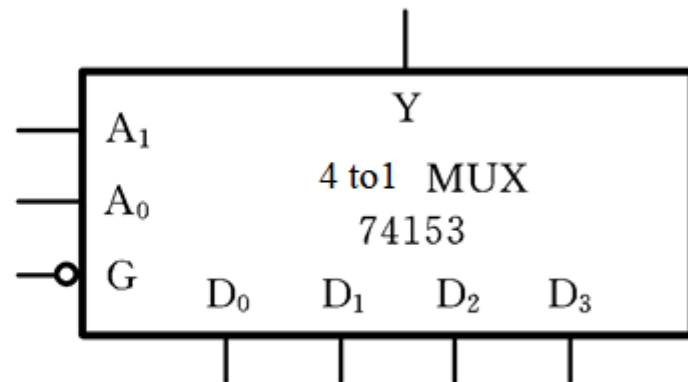
Common MSI multiplexers have 4-way, 8-way, and 16-way.

(1) Four way data multiplexers 74153

Figures (a) and (b) show the pin arrangement diagram and logical symbols of the dual 4-way multiplexers Model 74153. There are dual 4-way multiplexers in the chip. **where, $D_0 \sim D_3$ are data inputs; A_1 , A_0 are the selection inputs; Y is the output; G is the enable.**



(a)



(b)

(2) 74153 function table

74153 function table

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Control input $A_1 \quad A_0$		Data input $D_0 \quad D_1 \quad D_2 \quad D_3$				Output Y
0	0	D_0	d	d	d	D_0
0	1	d	D_1	d	d	D_1
1	0	d	d	D_2	d	D_2
1	1	d	d	d	D_3	D_3

(3) 74153 Output function

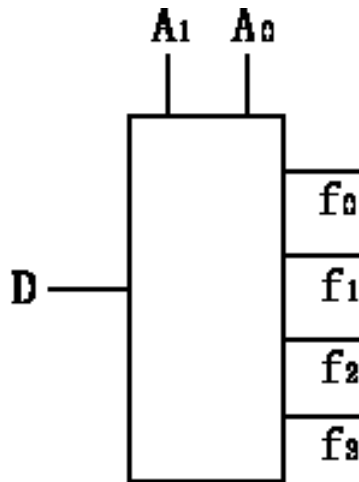
$$Y = \overline{A_1} \overline{A_0} D_0 + \overline{A_1} A_0 D_1 + A_1 \overline{A_0} D_2 + A_1 A_0 D_3 = \sum_{i=0}^3 m_i D_i$$

where, m_i are the control variables of the minterms of A_1, A_0 ,
 D_i is data input of the i th input, each value is 0 or 1.

II. Demultiplexer

Demultiplexer (also known as data distributor), commonly used DEMUX.

The structure of the demultiplexer is the opposite of that of the multiplexer. It is a single-input, multi-output combination logic component that determines the output from which the input is derived. As shown in the figure is the logical symbol of the 4-way splitter.



where, D is the data input; A₁, A₀ are the control input, $f_0 \sim f_3$ are the data output



4 way demultiplexer function table

Function table					
A_1	A_0	f_0	f_1	f_2	f_3
0	0	D	0	0	0
0	1	0	D	0	0
1	0	0	0	D	0
1	1	0	0	0	D

As seen from the function table, the output expression of the 4-way demultiplexer is

$$f_0 = \overline{A_1}\overline{A_0}D = m_0D \quad ; \quad f_1 = \overline{A_1}A_0D = m_1D$$

$$f_2 = A_1\overline{A_0}D = m_2D \quad ; \quad f_3 = A_1A_0D = m_3D$$

where, $m_i(i=0 \sim 3)$ are the minterms of the control input.





7.2 Common medium-scale sequential logic circuit

The most typical sequential logic circuits in digital systems are counters and registers.

7.2.1 Counters

1. What is a counter?

Generally speaking, a counter is a sequential logic circuit that can sequentially pass a predetermined state under the action of an input signal.

For commonly used integrated circuit counting products, it can be defined as follows:

Counter: A sequential logic circuit that counts input pulses. The **pulse signal** being counted is called a “count pulse”.





When the counter is running, the state it experiences is periodic and always circulates in a limited number of states. Usually, the total number of states included in one cycle is called the “Mode” of the counter.

2. Categories

There are many types of counters, and there are usually different classification methods.

- (1) According to its working mode, it can be divided into synchronous counter and asynchronous counter;
- (2) According to its carry system, it can be divided into binary counter, decimal counter and arbitrary counter;
- (3) According to its function, it can be divided into addition counter, subtraction counter and up/down reversible counter.

3. Function

Generally have functions such as counting, saving, resetting, presetting, etc..

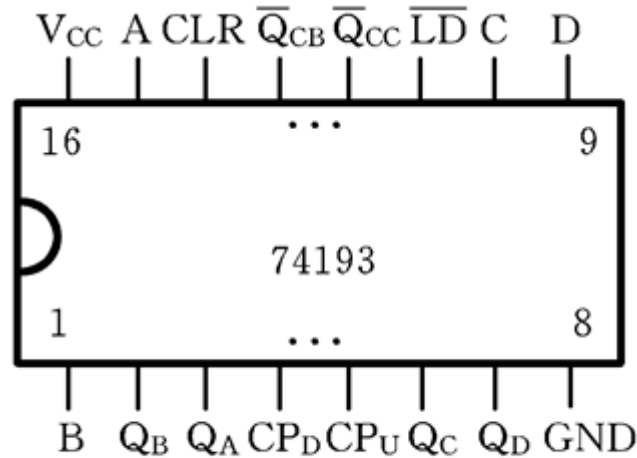


Typical chip

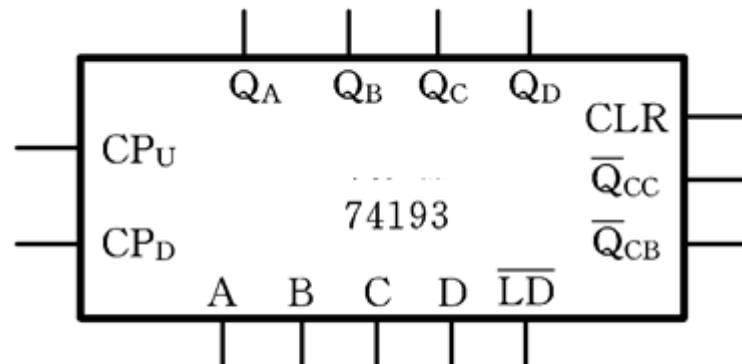
--- 4-bit binary synchronous reversible counter 74193

1. 74193 pin arrangement diagram and logical symbols

The 74193 pin arrangement diagram and logical symbols are shown in Figure (a) and (b), respectively.

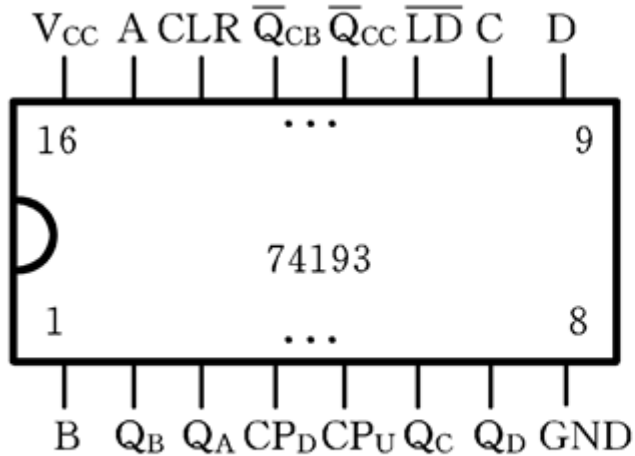


(a)

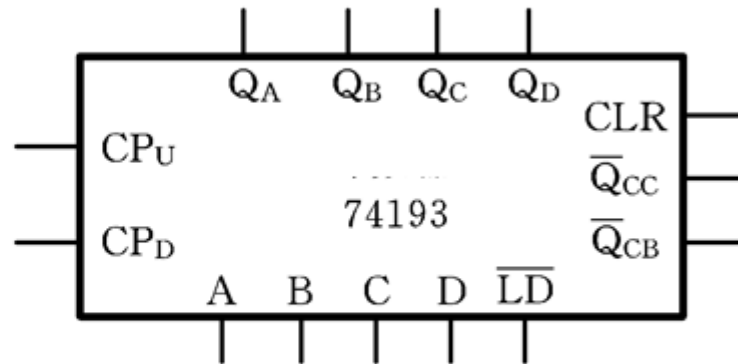


(b)

2. Pin Functions



(a)



(b)

CLR: clear

LD: Presetting control

D C B A: Presetting values

CPu: accumulate count pulse

CPd: Decrement count pulse

Q_{d-A} counting values

Q_{CC} carry

Q_{cb} borrow



7.2.2 Register

Register: A common logical component used in digital systems to store data or results.

Function: In addition to the basic functions of receiving data, saving data and transmitting data, the medium-sized integrated circuit registers usually have left and right shifts, serial, parallel input, serial, and output, and preset, clear and other functions. , is a multi-function register.

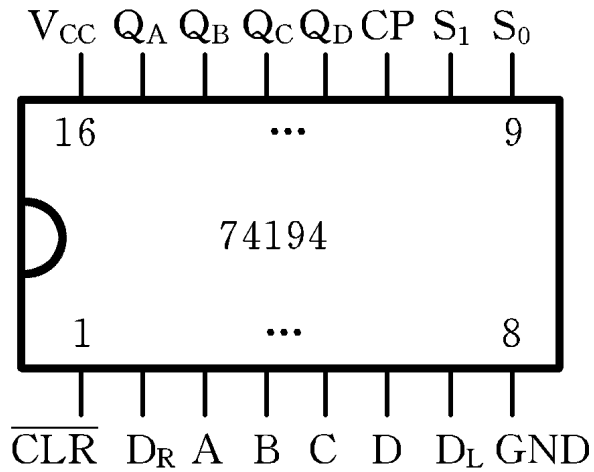
1. Typical chip

There are many types of medium scale integrated circuit registers. For example, the 74194 type is a commonly used 4-bit bidirectional register.

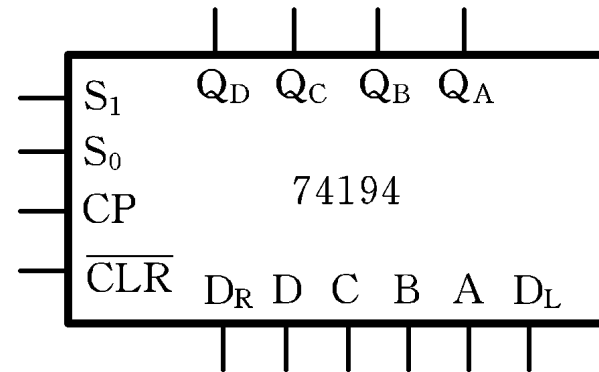


1. Pin arrangement diagram and logic symbol of 74194

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(a)



(b)

The 74194 has 10 inputs and 4 outputs.